Proposed course · graduate and undergraduate

Applied Agentic AI

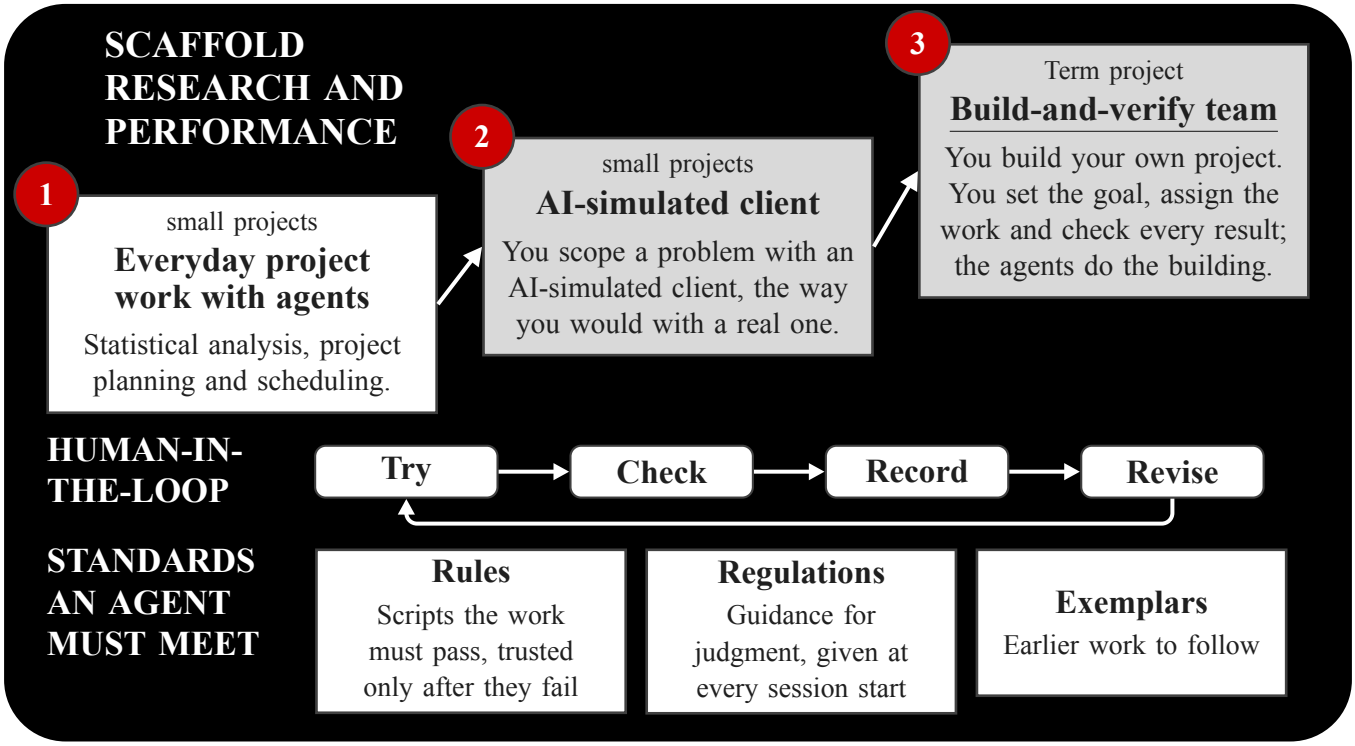
jbenhart44.github.io

John E. Benhart IV

Bring a real problem from your field, and learn to solve it with a team of AI agents that you direct.

You stay the human in the loop. AI should scaffold your work, not replace your judgment.

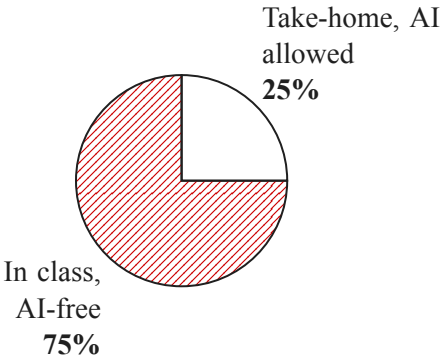
An interdisciplinary course to instruct students to utilize AI to help them reach new heights, whatever their background.



Council of Agents

Different roles give different answers, even from one model

Share of course grade



- Students use any AI assistant or coding agent, such as Claude or Claude Code, ChatGPT or Codex, Google Gemini, or GitHub Copilot.
- No programming course or prior AI experience is required.

Syllabus

jbenhart44.github.io/teaching/applied-agentic-ai-syllabus.pdf

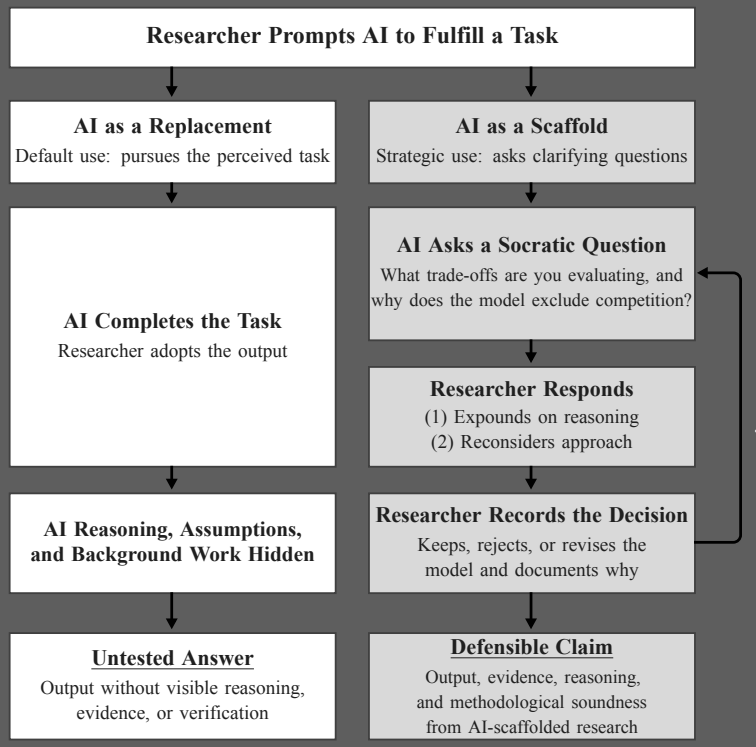
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AI-Centric: Efficiency

Offloading time-consuming/mundane tasks

- Summarizing literature
- Coding outlined algorithms
- Drafting text expeditiously
- Cleaning and analyzing data



Human-Centric: Novelty

Building on and extending existing literature

- Collecting new data
- Applying new principles from other disciplines into research
- Ideate on new phenomena
- Personal experiences shape research

Performance-Centric: AI-Scaffolded Academic Research

- AI directs researchers to most relevant research
- Leveraging coding LLMs to carry out well-defined tasks
- Parse down/combine complicated datasets
- Build minimal viable prototypes for potential research in days
- Researchers ideate and mediate what AI creates
- Iterating on task parameters and planning until prepared for execution

The goal in this course is to teach students how to amplify their research and learning through applied AI.